Provability facts for disjunction in the sequent calculus

The Open Logic Project contributors

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-first-order-logic-sequent-calculus/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the source uses the bussproofs doubleLine macro for stacked structural inferences and a parenthetical remark explaining it; delivered verbatim, doubleLine is provided by the bussproofs package on the host

Editorial note. The label prop:provability-lor names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own prop environment in the unit source.

Allowed standard prerequisites (editorial).

- Left and right disjunction rules. Left-or derives (A or B) on the left from A on the left and from B on the left; Right-or derives (A or B) on the right from A (or from B) on the right.. Item (1) combines the two Left-not derivations with a Left-or inference; item (2) uses Right-or on the initial sequents.

Proposition 0.1. *1. $A \vee B, \neg A, \neg B$ is inconsistent.*

2. Both $A \vdash A \vee B$ and $B \vdash A \vee B$.

Proof. 1. We give derivation of the sequent $A \vee B, \neg A, \neg B \Rightarrow$:

$$\dfrac{\dfrac{\dfrac{AA}{\neg A, A}}{A, \neg A, \neg B} \qquad \dfrac{\dfrac{BB}{\neg B, B}}{B, \neg A, \neg B}}{A \vee B, \neg A, \neg B}$$

(Recall that double inference lines indicate several weakening, contraction, and exchange inferences.)

2. Both sequents $A \Rightarrow A \vee B$ and $B \Rightarrow A \vee B$ have derivations:

$$\frac{AA}{AA \vee B} \qquad \frac{BB}{BA \vee B}$$

□